Bench Grinder

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Bench Grinder

What it is used for

A bench grinder uses rotating abrasive wheels to grind, sharpen, deburr, or shape suitable metal workpieces. Correct wheel selection, secure guards and tool rests, controlled contact, and proper eye protection help reduce wheel breakage, flying-particle, and entanglement hazards.

Main parts

- Electric motor drives the grinding spindle.
- Abrasive grinding wheels remove material from the workpiece.
- Wheel guards contain fragments and limit contact with the wheel.
- Adjustable tool rests support the workpiece during grinding.
- Spark deflectors direct sparks and debris away from the operator.
- Eye shields provide an additional barrier where fitted.
- On/off switch controls the motor.
- Mounting base secures the grinder to the bench.
- Wheel flanges and retaining nuts secure the wheels to the spindle.

Before you start

1. Only trained personnel should operate the bench grinder.
2. Inspect the grinder, power cable, switch, wheel guards, tool rests, and eye shields for damage.
3. Check each wheel for cracks, chips, moisture damage, or other defects; use the manufacturer's approved inspection method.
4. Confirm that the wheel type, diameter, bore, and maximum speed are compatible with the grinder.
5. Ensure wheel guards are secure and tool rests are adjusted close to the wheel according to the manual.
6. Check that spark deflectors and eye shields are correctly positioned and the bench grinder is securely mounted.
7. Wear safety glasses and a face shield where required; secure long hair and loose clothing, and remove jewellery.

How to use

1. Confirm the workpiece material is suitable for the installed grinding wheel.
2. Stand to one side of the wheel during startup and switch on the grinder.
3. Allow the wheel to reach operating speed and observe for abnormal vibration, noise, or wobble.
4. Hold the workpiece firmly with both hands where appropriate and rest it securely on the tool rest.

5. Bring the workpiece into gentle contact with the wheel using controlled pressure.
6. Move the workpiece across the appropriate wheel surface to avoid local overheating and uneven wear.
7. Keep fingers clear of the wheel and do not force the workpiece into the abrasive surface.
8. Cool the workpiece as needed using an appropriate method and avoid touching hot metal with bare hands.
9. Switch off the grinder and wait until the wheels stop completely before leaving the machine or making adjustments.
10. Remove debris with an approved brush or tool after the wheel has stopped.
11. Isolate electrical power before changing wheels, adjusting guards, or performing maintenance.

⚠ Safety

- Never use a cracked, damaged, incorrectly rated, or overspeed wheel.
- Do not grind on the side of a wheel unless it is specifically designed for side grinding.
- Keep guards and tool rests correctly installed and adjusted; never operate with missing safety components.
- Do not wear gloves, loose clothing, or jewellery that could become caught in rotating parts.
- Keep flammable materials away from sparks and direct sparks away from people and combustible substances.
- Never stop a wheel by pressing an object against it or by hand.

Care & storage

- Isolate power before cleaning dust and debris from the grinder and surrounding bench.
- Inspect wheels, guards, flanges, tool rests, and mounting bolts regularly.
- Replace worn or damaged wheels only with compatible wheels installed according to the manual.
- Keep ventilation openings clear and maintain the grinder according to the manufacturer's schedule.
- Remove the grinder from service if unusual vibration, wheel damage, or electrical faults occur.

Common mistakes to avoid

- Using a wheel with the wrong speed rating or intended application.
- Grinding with a loose tool rest or missing wheel guard.
- Applying excessive pressure and overheating the workpiece.
- Leaving the machine before the wheel has completely stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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